

Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau

TEAM ID	SWTID-2026-6767
PROJECT TITLE	Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau
MAX MARKS	3 MARKS

3.4 – Technology Stack

Hardware Requirements:

Component	Specification
Processor	Intel Core i5 Processor
RAM	16 GB
Storage	256 GB SSD
Display	Minimum 14-inch HD Display (Recommended)
Internet Connectivity	Broadband Connection for Data Access and Software Updates

Software Requirements:

Software	Purpose
Tableau Desktop	Data visualization and dashboard creation
XAMPP	Local server environment for database connectivity
MySQL Workbench	Database design and management
Python	Data preprocessing and analysis
PyCharm	Python development environment
Git	Version control and project management
Microsoft Office (MS Word, Excel, PowerPoint)	Documentation, report preparation, and presentation
Windows 10/11 (64-bit)	Operating System

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Technology Stack:

Layer	Technology Used
Programming Language	Python
Visualization Tool	Tableau Desktop
Database	MySQL
Database Management Tool	MySQL Workbench
Server Environment	XAMPP
IDE	PyCharm
Version Control	Git
Documentation Tools	Microsoft Office (MS Word, Excel, PowerPoint)
Data Format	CSV, Excel
Operating System	Windows 10/11

Brief Description of the Technology Stack

- **Python** is used for data cleaning, pre-processing, and analytical operations on electric vehicle datasets.
- **Tableau Desktop** enables interactive dashboards and visual analysis of EV charging patterns and range performance.
- **MySQL** stores and manages structured datasets related to electric vehicles.
- **MySQL Workbench** facilitates database modelling and query execution.
- **XAMPP** provides a local development environment and database connectivity support.
- **PyCharm** serves as the integrated development environment for Python scripting.
- **Git** manages source code versions and enables collaboration during development.

ALPHA
— INNOVATORS —
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